Chapter 3

Dataset

3.1 Novel BPAC Dataset Creation

3.1.1 Background

As established in Chapters 1 and 2, no existing Bangla corpus lets the same sentence be observed across regional accents at scale. To close this gap we built the Bangla Parallel Accent Corpus (BPAC): a corpus where all regional and standard speakers read an identical set of sentences, so that the accent variable is isolated from linguistic content by construction. This gives supervised methods the paired data they need, establishes a ground truth for accent conversion, and makes controlled evaluation possible, since a source and a reference can always be paired on the same sentence. The rest of this chapter describes how the corpus was designed, recorded, and organised.

3.1.2 Text Corpus Selection

To make the models robust across various contexts, the text corpus was curated to be linguistically diverse and phonetically balanced. We did this in two stages: broad sourcing, then diversity-driven selection.

3.1.2.0.1 Data sourcing. We collected raw text data (articles, news reports, and blogs) from online repositories covering 12 distinct domains to maximise vocabulary coverage: Bangladeshi affairs, business and economy, education, entertainment, health and wellness, international affairs, literature, news, politics, religion, science and technology, and sports.

3.1.2.0.2 Diversity optimisation. From this large pool, we selected the subset of sentences offering the highest phonetic and lexical diversity through an iterative process:

1. We generated 100 candidate sets, each containing 400 sentences.
2. For each candidate set, we computed the Word Diversity Score (the count of unique words).
3. We manually evaluated the top-ranking sets to ensure they contained natural, diverse utterances of varied length.
4. We selected the set with the highest number of unique words, so that models are trained on a rich distribution of Bangla phonemes.
5. From this final set, we selected the 250 most variable sentences, considering word diversity, punctuation, and sentence length together.

The result is a fixed 250-sentence Standard Bengali script that every participant reads, regardless of region.

3.1.3 Participant Recruitment

The corpus is designed to cover the major regional zones of Bangladesh. We categorised the regions into eight primary divisions: Sylhet, Chattogram (Chittagong), Dhaka, Faridpur, Khulna, Mymensingh, Rajshahi, and Rangpur. Participants were recruited so that each region is represented by speakers who natively carry that region's accent.

The final speaker counts per region are given in Table 3.1.

3.1.4 Participant Description

We aimed to collect, for each region, a minimum of three male and three female speakers; as Table 3.1 shows, most regions met or exceeded this target, while in Rangpur and Mymensingh we could recruit two male speakers. All speakers were aged between 19 and 25 years, and most were students at various universities of Bangladesh. The narrow age band was a practical outcome of recruiting students, but it also helps the corpus: age-related voice variation stays small, so it does not mix into the accent comparison.

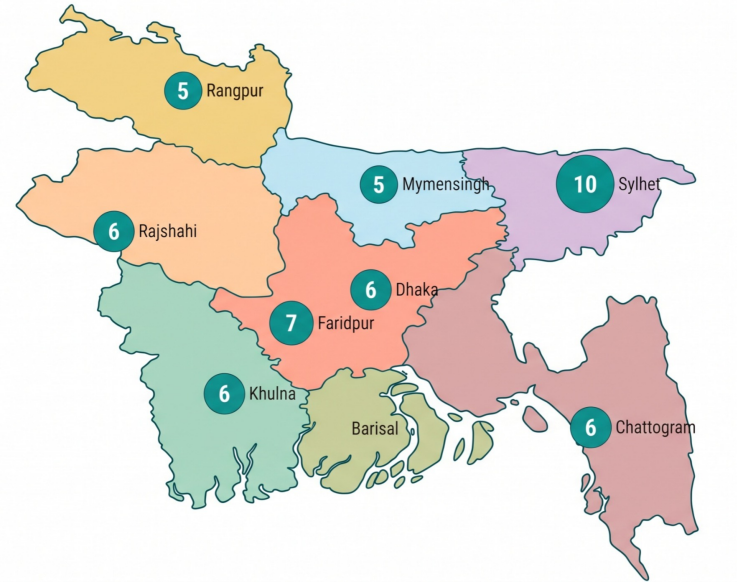


Figure 3.1: Map of Bangladesh illustrating the regions considered for participant recruitment.

Table 3.1: Participants per region.

Region	**Male**	**Female**	**Total**
Sylhet	6	4	10
Faridpur	4	3	7
Khulna	3	3	6
Chattogram (Chittagong)	3	3	6
Rangpur	2	3	5
Mymensingh	2	3	5
Rajshahi	3	3	6
Dhaka	3	3	6
Total	**26**	**25**	**51**

Recording logistics differed by gender, for a practical reason. Male speakers were recorded at the Audio Laboratory of Shahjalal University of Science and Technology (SUST). For female speakers, we found that many participants had issues, from both personal and social perspectives, with coming to the lab to record. Rather than lose their participation, we developed a mobile application so they could record on their own. Every female participant was first trained in an online meeting to ensure background noise stays minimal during recording, including simple but important habits like choosing a quiet room and turning the fan off.

3.1.5 Recording Setup and Specifications

3.1.5.1 Environment

3.1.5.1.1 Male participants. Recordings for male participants were conducted in the Audio Laboratory at SUST. This controlled, sound-proof environment ensures near-zero background noise and studio-quality capture.

3.1.5.1.2 Female participants. Audio was collected using SUST Kotha, a Flutter-based Android application we developed for speech corpus collection. Participants recorded the prompted sentences through a hold-to-record interface, and recordings were stored locally on-device and exported manually as ZIP archives. Importantly, no automatic gain control, noise suppression, echo cancellation, or any other audio preprocessing was applied by the app, so the captured signal is the raw microphone signal. We manually evaluated the collected audio and found the quality consistently high with minimal noise, which confirmed that the training given to the participants worked.

3.1.5.2 Technical Specifications

Male recordings were captured using high-fidelity digital recording software (Windows Sound Recorder) through an M-Track 2X2M high-quality audio interface. The specific parameters are detailed in Table 3.2; the corresponding specifications of the mobile-app pipeline used for female participants are given in Table 3.3.

These high-fidelity raw recordings allow effective downsampling (to 22.05 kHz or 16 kHz) as required by various model architectures (e.g., HuBERT-family encoders or vocoders) without loss of critical acoustic information. The raw corpus keeps the native formats of the two pipelines



Figure 3.2: Recording sessions at the SUST Audio Laboratory.

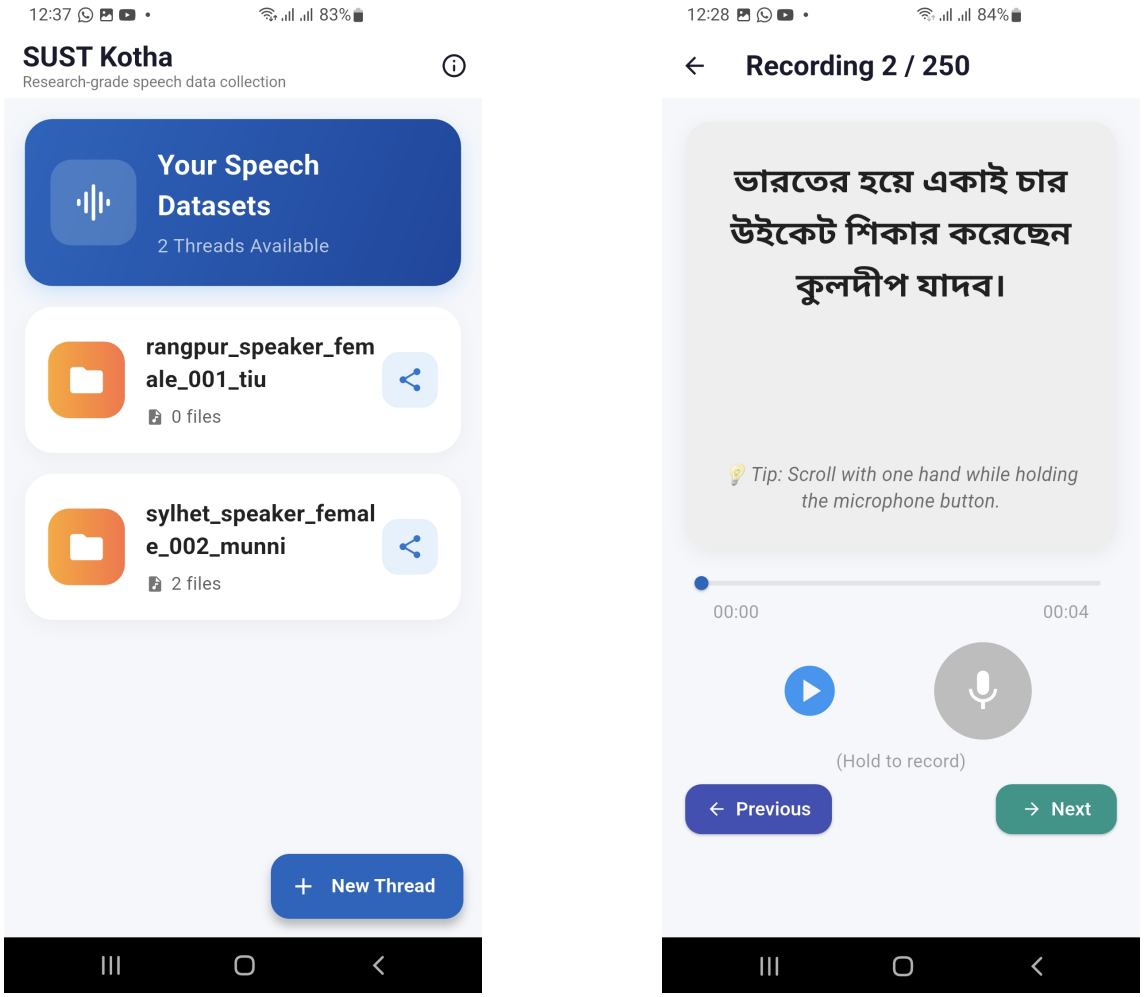


Figure 3.3: The SUST Kotha recording application.

Table 3.2: Technical specifications of the laboratory (male) recordings.

Parameter	**Specification**
Recording software	Windows Sound Recorder
Audio interface	M-Track 2X2M
Channels	2 (stereo)
Sample rate	44.1 kHz / 48 kHz
Bit depth	16-bit
Format	Uncompressed WAV (PCM)

Table 3.3: Technical specifications of the mobile-app (female) recordings.

Parameter	**Specification**
Recording application	SUST Kotha (Flutter, Android)
Recording interface	Hold-to-record prompted sentences
Channels	1 (mono)
Sample rate	44.1 kHz
Bit depth	16-bit PCM
Format	Uncompressed WAV
On-device processing	None (no AGC, noise suppression, or echo cancellation)
Storage / export	Local on-device storage, manual ZIP export

(stereo 44.1/48 kHz from the lab, mono 44.1 kHz from the app); format standardisation is treated as a preprocessing step of each downstream experiment rather than baked into the corpus itself.

3.1.6 BPAC Corpus Structure

The final corpus is structured as a matrix, which makes pairing for training and evaluation straightforward: for any sentence identifier, recordings of the same content are available from every speaker of every region. Table 3.4 illustrates the logical organisation with two regions as an example, one acting as the source accent and one as the conversion target.

Table 3.4: Logical structure of the BPAC dataset (illustrated with Sylhet as source and Faridpur as target).

Sentence ID	**Sylhet: Person 1**	**…**	**Sylhet: Person N**	**Faridpur: Person 1**	**…**	**Faridpur: Person N**
Sent_001	Audio_S1_001	…	Audio_SN_001	Audio_F1_001	…	Audio_FN_001
Sent_002	Audio_S1_002	…	Audio_SN_002	Audio_F1_002	…	Audio_FN_002
…	…	…	…	…	…	…
Sent_250	Audio_S1_250	…	Audio_SN_250	Audio_F1_250	…	Audio_FN_250

On disk, the corpus is organised as one folder per region, one subfolder per speaker, and one WAV file per sentence:

```
data/
|-- BPAC/
|   |-- chattogram/
```

```
|   |   |-- speaker_female_001_mahi/
|   |   |   |-- sound_001.wav
|   |   |   |-- sound_002.wav
|   |   |   |-- sound_003.wav
|   |   |   |-- sound_004.wav
|   |   |   |-- sound_005.wav
|   |   |   |-- sound_006.wav
|   |   |   |-- sound_007.wav
|   |   |   |-- ...
|   |-- dhaka/
|   |-- Faridpur/
|   |-- khulna/
|   |-- mymensingh/
|   |-- rajshahi/
|   |-- rangpur/
|   |-- sylhet/
```

Speaker folders follow the pattern `speaker_<gender>_<id>_<name>`, and audio files the pattern `sound_<NNN>.wav`, where `<NNN>` (001–250) is the sentence identifier linking each recording to its sentence in the script.

3.1.7 Corpus High-Level Details

The completed corpus contains 51 speakers across the eight regions, each having read all 250 sentences: 12,750 utterances in total, amounting to 27 hours 44 minutes 29 seconds (27.74 h) of audio. Durations reported here are raw file-header durations, including natural leading and trailing silence. The mean utterance length is 7.83 seconds, and per-speaker totals range from 23.0 to 48.3 minutes, so every speaker contributes a substantial amount of audio. Table 3.5 breaks the corpus down by region, Table 3.6 by gender, and Table 3.7 by region and gender jointly.

Speaker counts are moderately uneven across regions (five to ten per region, with Sylhet the largest at ten), and the gender balance is close (6,500 male versus 6,250 female utterances). We record both facts here because they matter later: the regional imbalance bounds the strength of the

Table 3.5: Audio volume per region.

Region	Total (min)	Mean utterance (s)	Utterances
Sylhet	297.18	7.13	2,500
Faridpur	231.14	7.92	1,750
Dhaka	210.20	8.41	1,500
Chattogram	206.72	8.27	1,500
Khulna	200.29	8.01	1,500
Rajshahi	190.30	7.61	1,500
Mymensingh	167.04	8.02	1,250
Rangpur	161.62	7.76	1,250

Table 3.6: Audio volume per gender.

Gender	Total (s)	Mean utterance (s)	Utterances
Male	53,095.1	8.17	6,500
Female	46,774.1	7.48	6,250

Table 3.7: Audio volume (minutes) per region and gender.

Region	Female	Male
Chattogram	89.53	117.18
Dhaka	92.33	117.87
Faridpur	100.31	130.83
Khulna	114.89	85.39
Mymensingh	99.61	67.44
Rajshahi	88.16	102.14
Rangpur	89.78	71.84
Sylhet	104.96	192.22

speaker–accent confound analysed in our exploratory data analysis, and the gender split matters because pitch statistics must be read within gender.

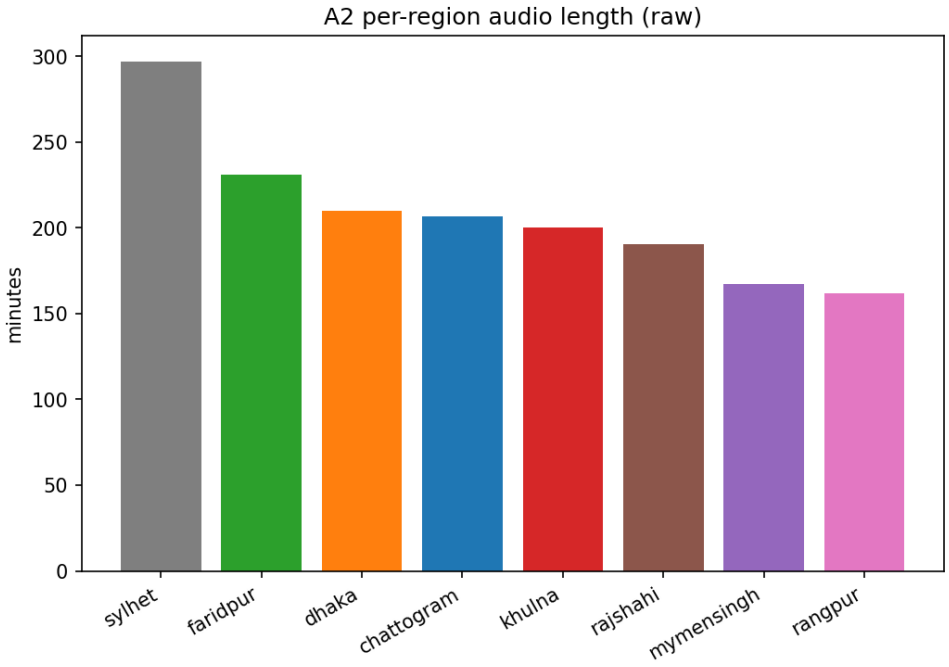


Figure 3.4: Total audio per region.

3.1.8 Participant Consent

All participants took part with informed consent and received compensation. For male participants, consent was taken on paper: each participant agreed to take part in the research by signing with their name, and compensation was provided to each of them. For female participants, consent was collected through a Google Form, and they also received compensation.

In both cases, the consent form clearly stated the purpose of the study and why the dataset was being collected, and before recording began, each participant clearly understood what their recordings would be used for.

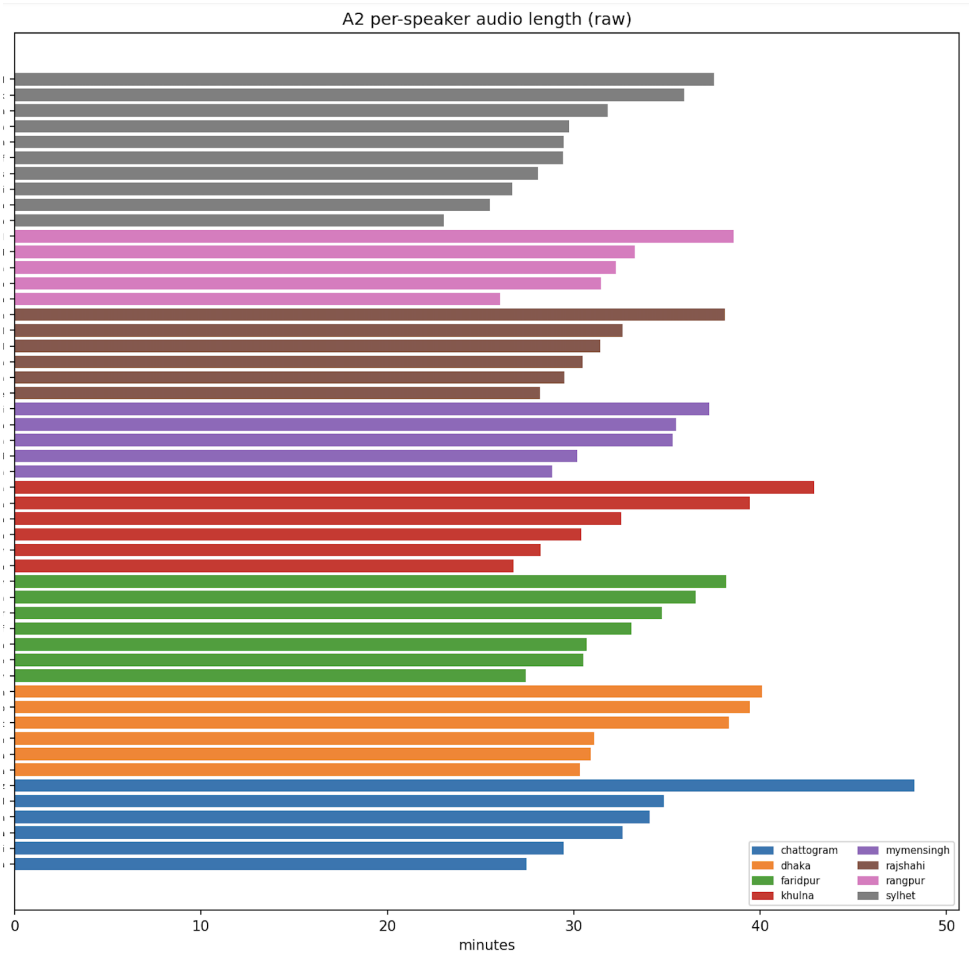
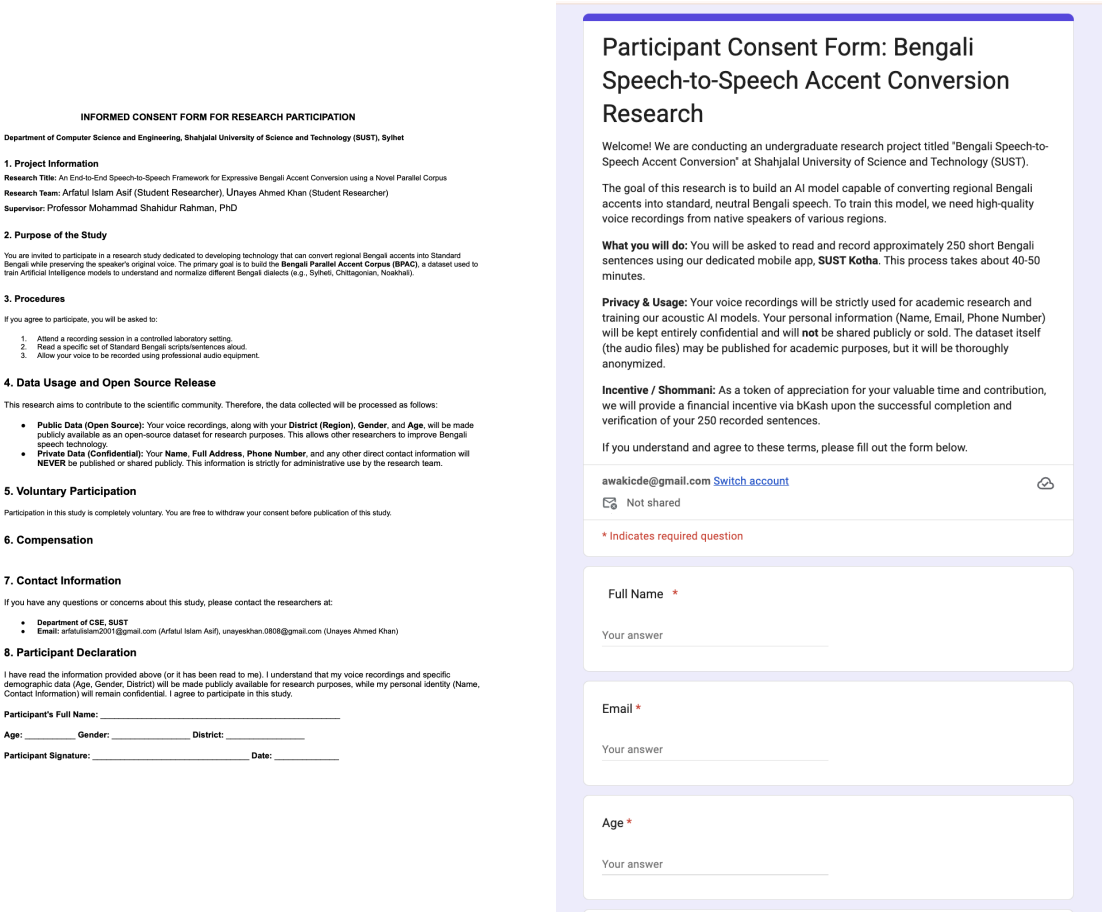


Figure 3.5: Total audio per speaker, coloured by region.



INFORMED CONSENT FORM FOR RESEARCH PARTICIPATION

Department of Computer Science and Engineering, Shahjalal University of Science and Technology (SUST), Sylhet

1. Project Information

Research Title: An End-to-End Speech-to-Speech Framework for Expressive Bengali Accent Conversion using a Novel Parallel Corpus

Research Team: Arfatul Islam Asif (Student Researcher), Unayes Ahmed Khan (Student Researcher)

Supervisor: Professor Mohammad Shahidur Rahman, PhD

2. Purpose of the Study

You are invited to participate in a research study dedicated to developing technology that can convert regional Bengali accents into Standard Bengali while preserving the speaker's original voice. The primary goal is to build the **Bengali Parallel Accent Corpus (BPAC)**, a dataset used to train Artificial Intelligence models to understand and normalize different Bengali dialects (e.g., Sylheti, Chittagonian, Noakhali).

3. Procedures

If you agree to participate, you will be asked to:

1. Attend a recording session in a controlled laboratory setting.
2. Read a specific set of Standard Bengali scripts/sentences aloud.
3. Allow your voice to be recorded using professional audio equipment.

4. Data Usage and Open Source Release

This research aims to contribute to the scientific community. Therefore, the data collected will be processed as follows:

- **Public Data (Open Source):** Your voice recordings, along with your **District (Region)**, **Gender**, and **Age**, will be made publicly available as an open-source dataset for research purposes. This allows other researchers to improve Bengali speech technology.
- **Private Data (Confidential):** Your **Name**, **Full Address**, **Phone Number**, and any other direct contact information will **NEVER** be published or shared publicly. This information is strictly for administrative use by the research team.

5. Voluntary Participation

Participation in this study is completely voluntary. You are free to withdraw your consent before publication of this study.

6. Compensation

7. Contact Information

If you have any questions or concerns about this study, please contact the researchers at:

- **Department of CSE, SUST**
- **Email:** arfatulislam2001@gmail.com (Arfatul Islam Asif), unayeskhan.0808@gmail.com (Unayes Ahmed Khan)

8. Participant Declaration

I have read the information provided above (or it has been read to me). I understand that my voice recordings and specific demographic data (Age, Gender, District) will be made publicly available for research purposes, while my personal identity (Name, Contact Information) will remain confidential. I agree to participate in this study.

Participant's Full Name: _____

Age: _____ **Gender:** _____ **District:** _____

Participant Signature: _____ **Date:** _____

Participant Consent Form: Bengali Speech-to-Speech Accent Conversion Research

Welcome! We are conducting an undergraduate research project titled "Bengali Speech-to-Speech Accent Conversion" at Shahjalal University of Science and Technology (SUST).

The goal of this research is to build an AI model capable of converting regional Bengali accents into standard, neutral Bengali speech. To train this model, we need high-quality voice recordings from native speakers of various regions.

What you will do: You will be asked to read and record approximately 250 short Bengali sentences using our dedicated mobile app, **SUST Kotha**. This process takes about 40-50 minutes.

Privacy & Usage: Your voice recordings will be strictly used for academic research and training our acoustic AI models. Your personal information (Name, Email, Phone Number) will be kept entirely confidential and will **not** be shared publicly or sold. The dataset itself (the audio files) may be published for academic purposes, but it will be thoroughly anonymized.

Incentive / Shommani: As a token of appreciation for your valuable time and contribution, we will provide a financial incentive via bKash upon the successful completion and verification of your 250 recorded sentences.

If you understand and agree to these terms, please fill out the form below.

awakicde@gmail.com Switch account

Not shared

* Indicates required question

Full Name *

Your answer

Email *

Your answer

Age *

Your answer

Figure 3.6: Consent documentation. Left: paper consent form used for male participants. Right: Google Form used for female participants.